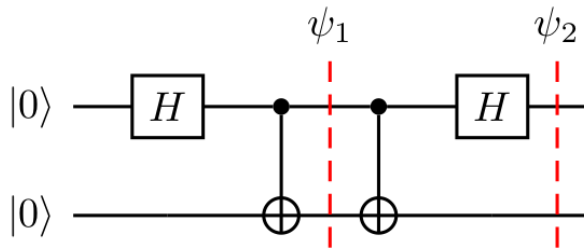


3.



i.

$$|0\rangle \xrightarrow{H} = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

After the CNOT gate is applied to the qubit:

$$\psi_1 = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

After using second CNOT again,

$$\psi_{II} = \text{CNOT } \psi_1 = \frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

After using Hadamard,

$$\psi_2 = \psi_{II} H = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |0\rangle + \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) |1\rangle$$

$$\frac{\left(\frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right)}{\sqrt{2}}$$

$$= \frac{1}{2} [|00\rangle + |10\rangle + |00\rangle - |10\rangle]$$

$$= |00\rangle$$

ii. At the end

2.

a.

Completed
Feb 10, 2024 2:30 PM (in 1.8s)

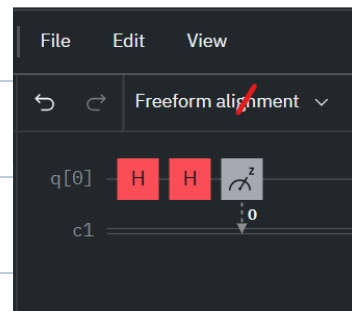
Compute resource
ibmq_qasm_simulator

Status timeline Completed

Details

Results

```
{
  "quasi_dists": [
    {
      "0": 1
    }
  ],
  "metadata": [
    {
      "shots": 1024,
      "circuit_metadata": {},
      "readout_mitigation_overhead": 1,
      "readout_mitigation_time": 0.0202
    }
  ]
}
```



7968 Setup and run

Qiskit Read only

Open in Quantum Lab

```
1 from qiskit import QuantumRegister, ClassicalRegister, QuantumCircuit
2 from numpy import pi
3
4 qreg_q = QuantumRegister(1, 'q')
5 creg_c = ClassicalRegister(1, 'c')
6 circuit = QuantumCircuit(qreg_q, creg_c)
7
8 circuit.h(qreg_q[0])
9 circuit.h(qreg_q[0])
10 circuit.measure(qreg_q[0], creg_c)
11 # @columns [0,1,2]
```

b.

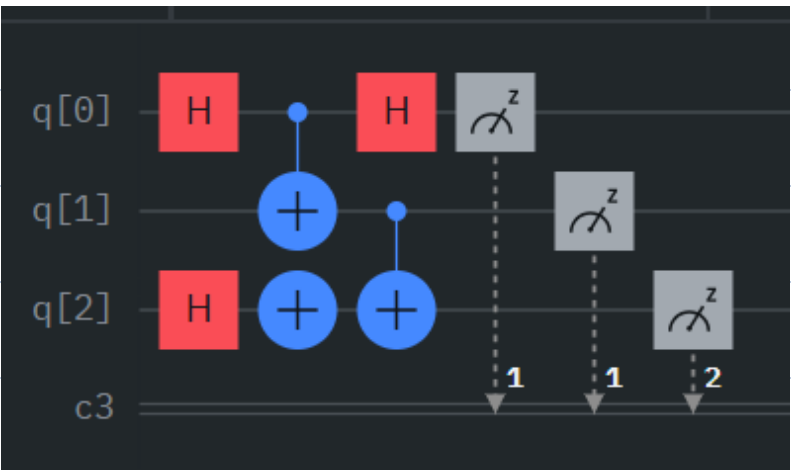
Compute resource
ibmq_qasm_simulator

Status timeline ✓ Completed ▾

Details ▾

Results ^

```
{
  "quasi_dists": [
    {
      "100": 0.2724609375,
      "110": 0.232421875,
      "000": 0.2451171875,
      "010": 0.25
    }
  ],
  "metadata": [
    {
      "shots": 1024,
      "circuit_metadata": {},
      "readout_mitigation_overhead": 1,
      "readout_mitigation_time": 0.0231
    }
  ]
}
```



```
1 from qiskit import
  QuantumRegister,
  ClassicalRegister,
  QuantumCircuit
2 from numpy import pi
3
4 qreg_q = QuantumRegister(3,
  'q')
5 creg_c = ClassicalRegister
  (3, 'c')
6 circuit = QuantumCircuit
  (qreg_q, creg_c)
7
8 circuit.h(qreg_q[0])
9 circuit.h(qreg_q[2])
10 circuit.cx(qreg_q[0], qreg_q
  [1])
11 circuit.x(qreg_q[2])
12 circuit.h(qreg_q[0])
13 circuit.cx(qreg_q[1], qreg_q
  [2])
14 circuit.measure(qreg_q[0],
  creg_c[1])
```

l. a.

i. $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

passing this state through another Hadamard gate,

$$H\left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \left[\left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right) \right]$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

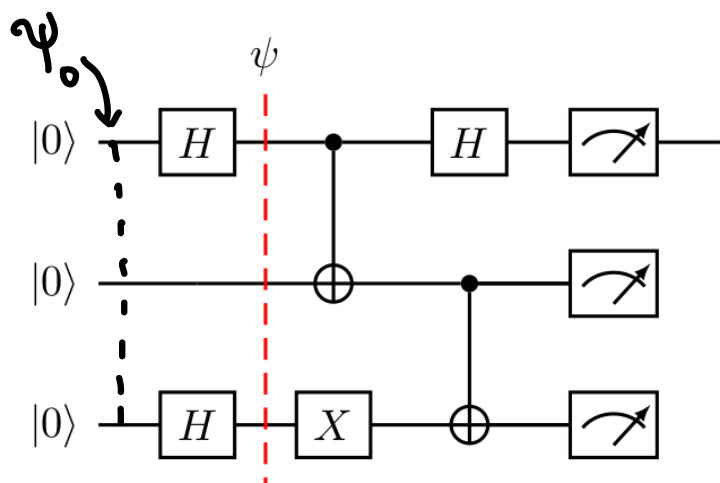
$$\text{ii } H * H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Using the $H * H$ with qubit state:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

ii



$$\psi_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$\psi =$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\psi = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \otimes \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{CNOT} \otimes X = \begin{bmatrix} 1 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} & 0[x] & 0[x] & 0[x] \\ 0[x] & 1[x] & 0[x] & 0[x] \\ 0[x] & 0[x] & 0[x] & 1[x] \\ 0[x] & 0[x] & 1[x] & 0[x] \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\left\{ \begin{array}{cc|cc|cc} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{array} \right\}$$

$$\frac{1}{2} \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

H ⊗ CNOT

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\left\{ \begin{array}{cc} 0001 & 0001 \\ 0010 & 0010 \\ 1000 & -1000 \\ 0100 & 0-100 \\ 0001 & 0001 \\ 0010 & 00-10 \end{array} \right\}$$

$$\frac{1}{\sqrt{2}} \left\{ \begin{array}{cc} 1000 & 1000 \\ 0100 & 0100 \\ 0001 & 0001 \\ 0010 & 0010 \\ 1000 & -1000 \\ 0100 & 0-100 \\ 0001 & 0001 \\ 0010 & 00-10 \end{array} \right\} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \frac{1}{2\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$$

ii Ket notation

$$\psi_0 = |000\rangle$$

$$\psi = H(|0\rangle) \otimes |0\rangle \otimes H(|0\rangle)$$

$$= \left\{ \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right\} \otimes |0\rangle \otimes \left\{ \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right\}$$

$$= \left\{ \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |10\rangle \right\} \otimes \left\{ \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \right\}$$

$$= \frac{1}{2} |000\rangle + \frac{1}{2} |001\rangle + \frac{1}{2} |100\rangle + \frac{1}{2} |101\rangle$$

$$CNOT_{01} \times_2 \left(\frac{1}{2} |000\rangle + \frac{1}{2} |001\rangle + \frac{1}{2} |100\rangle + \frac{1}{2} |101\rangle \right)$$

$$= \frac{1}{2} |001\rangle + \frac{1}{2} |000\rangle + \frac{1}{2} |111\rangle + \frac{1}{2} |110\rangle$$

$$H_0 \text{ CNOT}_{1,2} \left(\frac{1}{2} |001\rangle + \frac{1}{2} |000\rangle + \frac{1}{2} |111\rangle + \frac{1}{2} |110\rangle \right)$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} \left\{ |001\rangle + |101\rangle + |000\rangle + |100\rangle + |010\rangle - |110\rangle + |011\rangle - |111\rangle \right\}$$

$$= \frac{1}{2\sqrt{2}} \left\{ |001\rangle + |101\rangle + |000\rangle + |100\rangle + |010\rangle - |110\rangle + |011\rangle - |111\rangle \right\}$$

3ii The qubits are entangled after Ψ_1 as they cannot be factorized into two different single qubit states.

To check mathematically

$$AD = BC$$

$$\text{for } A|00\rangle + B|01\rangle + C|10\rangle + D|11\rangle$$

$$\frac{1}{2} |00\rangle \quad 0 |01\rangle \quad 0 |10\rangle \quad \frac{1}{2} |11\rangle$$

$$\frac{1}{2} \times \frac{1}{2} \neq 0 \quad \Rightarrow \quad \frac{1}{4} \neq 0$$

The qubits are entangled in ψ_1 .